

Algebra II with Trigonometry

Chapter 6.4

Mental Math (GPT-5.4)

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[4533650_alg-2trig-h-chp-6-4-note.pdf](#)

- 1. Basic Periods** What is the period of $y = \tan x$, $y = \cot x$, $y = \sec x$, and $y = \csc x$?
- 2. Tangent Asymptotes** For $y = \tan x$, where are the vertical asymptotes?
- 3. Cotangent Asymptotes** For $y = \cot x$, where are the vertical asymptotes?
- 4. Reciprocal Identity Check** Which two trig functions are graphed by using the identities $\csc x = \frac{1}{\sin x}$ and $\sec x = \frac{1}{\cos x}$?
- 5. Cosecant Asymptotes** At what x -values does $y = \csc x$ have vertical asymptotes?
- 6. Secant Asymptotes** At what x -values does $y = \sec x$ have vertical asymptotes?
- 7. Period from a Transformation** What is the period of $y = 3 \tan(2x)$?
- 8. Shifted Tangent Interval** Give an appropriate interval for one period of $y = \tan\left(x - \frac{\pi}{4}\right)$.
- 9. Cotangent Transformation** What is the period of $y = 2 \cot(5x)$?
- 10. Secant/Cosecant Transformation** What is the period of $y = 4 \sec(3x)$, and what is one appropriate interval for one complete period?